Fiscal Year:	FY 2007	**Task Last Updated:**	FY 03/09/2007
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Project Title:	Microgravity Effects on In Vivo Transvascular Transport and Vascular Control		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Cellular biotechnology & tissue engineering		
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Comments:			
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No. of PhD Candidates:		**No. of Master' Degrees:**	
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Flight Assignment:	NOTE: Received NCE to 2/28/2007, per N. Hall (9/06)		
Key Personnel Changes/Previous PI:	NOTE: Originally John Tarbell's grant with Norman Harris as CoI. Harris now PI for the in vivo studies (his portion of the research).		
COI Name (Institution):			
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	When astronauts are exposed to a microgravity environment there is a characteristic tissue fluid shift from the lower extremities to the upper extremities (cephalad fluid shift), which is reversed during re-entry to normal gravity with concomitant reduction in orthostatic tolerance. The biofluid mechanical and biotransport mechanisms which control this fluid shift phenomena are not well understood and will be addressed in the proposed research. A major fraction of the fluid shift occurs by exchange of water and solutes across the endothelial cell (EC) layer which lines all blood vessels and provides the principal barrier to transport of material between the blood and the underlying tissue. Major alterations in vascular pressure associated with removal (or addition) of the gravitational field changes the transvascular pressure differential, which drives fluid transport between the blood and tissue compartments. We have shown, using a tissue culture model of the EC transport barrier in vitro, that a sudden increase in transvascular water flux induces an early		

Task Description:

adaptive response in which the hydraulic conductivity (water flux / pressure differential) decreases over a one-hour period. This mechanism tends to maintain constant transmural flow in the presence of changing vascular pressure conditions. However, these in vitro experiments have further revealed that beyond a one hour exposure to increased pressure, the hydraulic conductivity begins to increase, and reaches a new steady state level which is significantly elevated after about five hours. Furthermore, this increased hydraulic conductivity can be reversed by pharmacological intervention (inhibition of nitric oxide). We hypothesize that this alteration in hydraulic conductivity in response to altered pressure in microgravity contributes to the fluid shift problem. In the proposed research we will study the influence of altered vascular pressure on hydraulic conductivity and albumin permeability of endothelial layers using both in vivo and in vitro models and will explore the biomolecular mechanisms underlying these changes (specific objective 1).

A second important mechanism which controls transvascular fluid exchange is the regulation of blood flow to the tissue experiencing a change in blood pressure. For example, normal control of fluid transport to leg tissue after standing (which increases blood pressure due to the added gravitational force) is associated with the contraction of arterioles in response to increased pressure (the so-called “myogenic response”). The myogenic response limits blood flow and reduces downstream pressure and available surface area, all of which tend to restrict transvascular exchange. This mechanism prevents significant leg edema when we stand. The mechanical factors which drive the myogenic response, however, are not well understood. The traditional view has been that increases in smooth muscle cell (SMC) tension in the blood vessel wall driven by increases in pressure stimulate SMC contraction. However, using an in vitro SMC cell culture model, we have recently demonstrated that fluid flow shear stress on SMC can also drive significant cellular contraction. This strongly suggests that transmural flow, which imposes fluid shear stress on SMC and is driven by changes in vascular pressure as described above, will affect SMC contraction and the myogenic response. We therefore hypothesize that alterations in the myogenic response driven by changes in transvascular flow in altered gravity affect the normal mechanisms which regulate fluid accumulation in tissues. In the proposed research we will study the influence of altered vascular pressure on SMC tone using both in vivo and in vitro models. We will isolate the contributions of vascular tension and fluid flow shear stress to the overall myogenic response and elucidate the underlying biomolecular mechanisms (specific objective 2). Specific Objective 1. To determine the effects of changes in transvascular pressure on the transport of water and albumin across the endothelial layer. We will use in vivo and in vitro models to approach this specific objective. The in vivo studies will rely upon the microcirculation of the rat mesentery as the experimental model. Chronic pressure changes will be introduced using a micro-perfusion technique and hydraulic conductivity and albumin permeability will be measured over a 6 hour period using the modified Landis technique and fluorescent tracer intensity monitoring. The in vitro studies will employ bovine aortic endothelial cells (BAECs) and bovine retinal microvascular endothelial cells (BRECs) grown on porous supports as models of the endothelial transport barrier. Using a unique apparatus developed in our laboratory, we will monitor the water flux and albumin flux across the endothelial monolayer simultaneously for 6 hours after introducing a change in transendothelial pressure. Changes in the tight junctions and adherens junctions between endothelial cells, which are hypothesized to regulate transendothelial transport, will be monitored using antibodies to the tight junction proteins ZO-1 and occludin and the adherens junction protein, VE-cadherin. Signal transduction pathways involved in remodeling the tight junctions in response to changes in transmural flow (pressure) will then be elucidated.

Specific Objective 2. To determine the effects of changes in transvascular pressure on smooth muscle cell contraction. This specific aim will also be approached using both an in vivo and an in vitro model. The in vivo studies will determine the mechanisms by which myogenic tone is developed in the arterioles of the rat mesentery after sudden (step) and gradual (ramp) changes in pressure. Particular emphasis will be placed on determining the role of transmural interstitial flow in generating myogenic tone. For the first time, we will measure transmural flow during the myogenic response to determine whether it is regulated. In addition we will manipulate transmural flow at a fixed level of pressure change using osmotic agents in order to dissect out, for the first time, the contribution of transmural flow to the development of myogenic tone. The in vitro experiments will employ rat aortic SMC in a 3-dimensional collagen gel model through which controlled interstitial flow will be perfused. Contraction will be observed and quantified under a microscope using video microscopy and image processing, and the biochemical mechanisms mediating the response will be probed with a variety of chemical agonists and inhibitors. The proposed studies will provide direct understanding of the biotransport and biomolecular mechanisms which control transvascular fluid and solute exchange in response to changes in blood pressure in altered gravity environments. The studies can be extended to humans on the ground or in space and could ultimately lead to effective countermeasures to fluid shifts which impair physiological function in microgravity and upon return to normal gravity.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The research can be applied to situations on earth in which microvascular pressure and perfusion are altered (for example, when a person stands up after being in a prone position). Data collected from the research detail the cellular responses of microvascular endothelial and smooth muscle cells in response to changes in fluid shear stress and hydrostatic pressure.

This grant was funded as a joint project between Drs. John M. Tarbell and Norman R. Harris at the time when both investigators were faculty at The Pennsylvania State University. During the completion of the work, Dr. Tarbell relocated to City College of New York, and Dr. Harris relocated to LSU Health Sciences Center in Shreveport. The grant funded two major aspects of investigation: 1) in vitro (Tarbell) and 2) in vivo (Harris) determination of the effects of altered flow and hydrostatic pressure on transvascular transport and vascular control. This final report (which includes excerpts from the referenced publications) describes the approximate one-half of the total grant that was completed in the lab of Dr. Harris.

Specific Objective 1. To determine the effects of changes in transvascular pressure on the transport of water and albumin across the endothelial layer.

In response to an increase in intravascular pressure, arteriolar blood vessels will stretch initially, but then constrict in a second phase, which is called the myogenic response. Mechanisms of the myogenic response have not been completely established. We hypothesized that transvascular fluid filtration from plasma across smooth muscle cells and into the surrounding interstitium helps regulate arteriolar myogenic tone, or the state of arteriolar constriction. In our work,

Task Progress:	arteriolar diameters in the rat mesentery were monitored prior to and following vascular occlusion with a glass micropipette. Arteriolar occlusion not only gave an increase in hydrostatic pressure that initiated myogenic constriction upstream of the pipette, but also allowed measurement of fluid filtration rate by monitoring the movement of vascular red blood cells. A statistically significant correlation ($p<0.001$) existed between basal myogenic tone and fluid filtration. Additionally, the myogenic response was attenuated by 47 +/- 7% (N=10) when an osmotic solution of albumin, or albumin plus Ficoll, was infused into the bloodstream to decrease fluid filtration by 53 +/- 3%. Moreover, the same inhibition of myogenic tone was found in isolated, cannulated rat soleus muscle arterioles when filtration was osmotically attenuated by intravascular dextran. Taken together, these results are consistent with the hypothesis that shear stress on arteriolar smooth muscle, induced by transvascular fluid filtration, is a contributing factor that helps control myogenic tone. Specific Objective 2. To determine the effects of changes in transvascular pressure on smooth muscle cell contraction. The effects of mechanical perturbations (shear stress, pressure) on microvascular permeability primarily have been examined in micropipette-cannulated vessels or in endothelial monolayers in vitro. The objective of this study was to determine whether acute changes in blood flow shear stress might influence measurements of hydraulic conductivity (Lp; permeability of the microvascular wall to water) in autoperfused microvessels in vivo. Rat mesenteric microvessels were observed via intravital microscopy. Occlusion of a third-order arteriole with a micropipette was used to divert and increase flow through a non-occluded capillary or fourth-order arteriolar branch. Transvascular fluid filtration rate in the branching vessel was measured with a Landis technique in which the declining distance between intravascular red blood cells was used to calculate transvascular filtration. Flow (shear)-induced increases in Lp disappeared within 20-30 s of the removal of the shear and could be eliminated with nitric oxide synthase inhibition. The shear-induced increase in Lp was greater in capillaries compared to terminal arterioles. An acute change in shear may regulate Lp by a nitric oxide-dependent mechanism that displays heterogeneity within a microvascular network. The final study addressed the effect of a sustained change in pressure on microvascular permeability assessed by hydraulic conductivity (Lp) measurements from microvessels of the rat mesentery. Using a micro-perfusion technique, transvascular filtration (normalized to surface area; Jv/S) and Lp were measured in small arterioles (baseline Lp = 0.26 x 10-7 cm/s/cmH2O) and venules (baseline Lp = 2.88 x 10-7 cm/s/cmH2O). The main finding of this study was that step increases in microvascular pressure led to time-dependent alterations of Lp. Immediately after a two-fold step increase in pressure, Jv/S increased in proportion to the pressure change. This observation is consistent with Starling's Law that predicts filtration proportional to the overall pressure gradient when Lp is constant. However, when Jv/S measurements continued for 60-90 minutes past the step in pressure, there was an initial decrease in Jv/S for 30 min ("sealing effect") followed by a substantial increase in Jv/S out to 90 min. The sustained increase in Jv/S suggests an increase in Lp of 36 +/- 7% for small arterioles and 42 +/- 5% for small venules. In addition, the increase in Lp in response to an increase in pressure was attenuated significantly by nitric oxide synthase (NOS) inhibition. These results indicate that a pressure-induced mechanical stimulus (possibly Jv) activates a nitric oxide-dependent biochemical response that leads to an increase in hydraulic conductivity.
Bibliography Type:	Description: (Last Updated: 03/13/2007)
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